```
Given a question and a candidate answer to the question, please extract a KG from the candidate
answer condition on the question and represent the KG with triples formatted with ("subject",
"predicate", "object"), each triplet in a line.
Please note that this is an EXTRACTION task, so DO NOT care about whether the content of the
candidate answer is factual or not, just extract the triplets from it.

Here are some in-context examples:

### Question:
Given these paragraphs about the Tesla bot, what is its alias?

### Candidate Answer:
Optimus (or Tesla Bot) is a robotic humanoid under development by Tesla, Inc. It was announced
at the company's Artificial Intelligence (AI) Day event on August 19, 2021.

### KG:
("Optimus", "is", "robotic humanoid")
("Optimus", "under development by", "Tesla, Inc.")
("Optimus", "also known as", "Tesla Bot")
("Tesla, Inc.", "announced", "Optimus")
("Announcement of Optimus", "occurred at", "Artificial Intelligence (AI) Day event")
("Artificial Intelligence (AI) Day event", "held on", "August 19, 2021")
("Artificial Intelligence (AI) Day event", "organized by", "Tesla, Inc.")

### Question:
here is some text about Andre Weiss, how many years was Andre at University of Dijon in Paris?

### Candidate Answer:
11 years

### KG:
("Andre Weiss at University of Dijon in Paris", "duration", "11 years")

Now generate the KG for the following candidate answer based on the provided question:

### Question:
{q}

### Candidate Answer:
{a}

### KG:
```

Figure 15: The prompt used for the GPT-4 and Claude 2 extractors. It requires a question and response from the LLM, and is provided with two in-context examples.

of RefChecker outperform the baselines with large margins across all the five metrics.

- RefChecker is more effective with a GPT-4 checker. The best results are achieved with a GPT-4 checker indicating that the main bottleneck lies in the checking module. In spite of that, RefChecker can still outperforms the baselines with a smaller checker AlignScore.
- Purely open-sourced combinations can also outperform the baselines which are using proprietary LLMs for both extractor and checker.

### C.2 Comparison on the SelfCheckGPT Dataset

We also ran REFCHECKER on the SelfCheckGPT dataset which contains 237 examples on WikiBio domain. The results are shown in Table 13. We can observe that 11 out of the 15 combinations (73%) of REFCHECKER outperform SelfCheckGPT.

## D Analysis of Internal Knowledge Bias

In this section, we further analyze the emergence of the hallucination from the perspective of the LLMs' bias to the internal knowledge. We analyze whether the evaluated model and the checker generate response based on their internal knowledge in Section D.1 and D.2, respectively. In general, we observe that LLMs/Checkers may incorporate internal knowledge even when provided with contextual information, contributing to the occurrence of hallucination.

| Zero Context Setting | | | | | |
|---|---|---|---|---|---|
| | Accuracy | Fact. F1 | Non-Fact. F1 | Pearson | Spearman |
| SelfCheckGPT | 77.99 | 54.03 | 85.53 | 35.40 | 43.15 |
| FActScore | 66.41 | 49.42 | 74.86 | 42.58 | 45.60 |
| FacTool | 84.94 | 73.29 | 89.52 | 59.78 | 62.57 |
| REFCHECKER | | | | | |
| *GPT-4 + GPT-4* | **93.82** | **86.89** | **95.96** | **83.95** | 82.35 |
| *GPT-4 + Claude 2* | 91.31 | 82.76 | 94.19 | 80.11 | 79.53 |
| *GPT-4 + NLI* | 83.98 | 71.28 | 88.89 | 60.81 | 62.32 |
| *GPT-4 + AlignScore* | 90.54 | 78.97 | 93.90 | 71.95 | 70.37 |
| *GPT-4 + RepC* | 89.96 | 81.16 | 93.16 | 77.42 | 77.26 |
| *Claude 2 + GPT-4* | 92.66 | 84.30 | 95.21 | 83.69 | **82.99** |
| *Claude 2 + Claude 2* | 92.08 | 83.40 | 94.8 | 80.04 | 79.39 |
| *Claude 2 + NLI* | 83.20 | 70.51 | 88.26 | 59.25 | 60.39 |
| *Claude 2 + AlignScore* | 90.54 | 78.97 | 93.90 | 75.07 | 73.83 |
| *Claude 2 + RepC* | 89.58 | 80.71 | 92.86 | 76.34 | 76.24 |
| *Mistral-SFT + GPT-4* | 92.47 | 83.40 | 95.13 | 80.88 | 78.88 |
| *Mistral-SFT + Claude 2* | 90.93 | 80.66 | 94.07 | 77.98 | 77.04 |
| *Mistral-SFT + NLI* | 89.96 | 78.86 | 93.42 | 72.89 | 72.07 |
| *Mistral-SFT + AlignScore* | **90.54** | 78.79 | **93.91** | 75.81 | 74.16 |
| *Mistral-SFT + RepC* | 89.38 | **80.43** | 92.72 | **77.14** | **76.74** |

Table 10: A comparison of RefChecker with previous works on our benchmark under Zero Context setting. We highlight the best results using proprietary LLMs with blue colors and best results results using pure open-source models with orange colors.

| Noisy Context Setting | | | | | |
|---|---|---|---|---|---|
| | Accuracy | Fact. F1 | Non-Fact. F1 | Pearson | Spearman |
| SelfCheckGPT | 58.55 | 51.63 | 63.74 | 36.31 | 32.15 |
| FActScore | 63.57 | 69.94 | 53.77 | 33.36 | 29.91 |
| FacTool | 68.40 | 72.84 | 62.22 | 46.35 | 38.69 |
| REFCHECKER | | | | | |
| *GPT-4 + GPT-4* | 74.54 | 76.42 | 72.32 | 64.56 | 57.30 |
| *GPT-4 + Claude 2* | 66.73 | 67.16 | 66.29 | 52.40 | 42.90 |
| *GPT-4 + NLI* | 66.73 | 74.54 | 52.01 | 39.69 | 32.98 |
| *GPT-4 + AlignScore* | 67.66 | 73.48 | 58.57 | 44.31 | 37.58 |
| *GPT-4 + RepC* | 65.99 | 74.04 | 50.67 | 28.19 | 28.94 |
| *Claude 2 + GPT-4* | 71.38 | 73.72 | 68.57 | 53.14 | 47.89 |
| *Claude 2 + Claude 2* | 63.38 | 63.59 | 63.18 | 39.96 | 34.08 |
| *Claude 2 + NLI* | 64.13 | 71.58 | 51.39 | 29.04 | 25.50 |
| *Claude 2 + AlignScore* | 69.70 | 74.33 | 63.04 | 46.39 | 41.26 |
| *Claude 2 + RepC* | 65.80 | 74.01 | 50.00 | 32.30 | 29.03 |
| *Mistral-SFT + GPT-4* | **75.28** | **77.57** | **72.46** | **67.29** | **59.94** |
| *Mistral-SFT + Claude 2* | 64.50 | 61.10 | 67.35 | 56.14 | 47.02 |
| *Mistral-SFT + NLI* | **70.82** | **75.12** | **64.72** | 52.21 | **45.61** |
| *Mistral-SFT + AlignScore* | 69.70 | 74.73 | 62.18 | **53.88** | 45.09 |
| *Mistral-SFT + RepC* | 65.99 | 73.97 | 50.94 | 38.11 | 31.01 |

Table 11: A comparison of RefChecker with previous works on our benchmark under Noisy Context setting. We highlight the best results using proprietary LLMs with blue colors and best results results using pure open-source models with orange colors.

| | Accurate Context Setting | | | | |
|---|---|---|---|---|---|
| | Accuracy | Fact. F1 | Non-Fact. F1 | Pearson | Spearman |
| SelfCheckGPT | 62.15 | 68.70 | 52.12 | 40.23 | 32.55 |
| FActScore | 69.37 | 78.57 | 46.30 | 27.80 | 27.05 |
| FacTool | 72.53 | 80.98 | 50.63 | 31.41 | 32.82 |
| REFCHECKER | | | | | |
| *GPT-4 + GPT-4* | 80.81 | 86.85 | 64.50 | 58.61 | 55.50 |
| *GPT-4 + Claude 2* | 77.46 | 84.04 | 61.68 | 48.62 | 48.66 |
| *GPT-4 + NLI* | 73.06 | 82.23 | 44.36 | 28.59 | 30.14 |
| *GPT-4 + AlignScore* | 76.23 | 83.44 | 57.94 | 49.97 | 46.89 |
| *GPT-4 + RepC* | 76.94 | 84.86 | 51.66 | 45.58 | 41.01 |
| *Claude 2 + GPT-4* | **82.22** | **87.64** | **68.34** | **60.99** | **58.96** |
| *Claude 2 + Claude 2* | 74.47 | 81.62 | 58.21 | 49.81 | 44.72 |
| *Claude 2 + NLI* | 72.36 | 81.68 | 43.73 | 27.39 | 29.12 |
| *Claude 2 + AlignScore* | 74.30 | 81.57 | 57.56 | 50.17 | 44.59 |
| *Claude 2 + RepC* | 77.46 | 85.25 | 52.24 | 52.11 | 42.78 |
| *Mistral-SFT + GPT-4* | 79.75 | 85.96 | 63.72 | 56.09 | 53.72 |
| *Mistral-SFT + Claude 2* | 70.95 | 77.85 | 57.80 | 38.82 | 40.75 |
| *Mistral-SFT + NLI* | 73.59 | 81.44 | 54.27 | 44.34 | 40.81 |
| *Mistral-SFT + AlignScore* | **74.12** | 81.60 | **56.38** | **46.34** | **43.22** |
| *Mistral-SFT + RepC* | 73.94 | **82.83** | 45.99 | 39.59 | 33.86 |

Table 12: A comparison of RefChecker with previous works on our benchmark under Accurate Context setting. We highlight the best results using proprietary LLMs with blue colors and best results results using pure open-source models with orange colors.

| | Pearson | Spearman |
|---|---|---|
| SelfCheckGPT | 78.32 | 78.30 |
| **RefChecker** | | |
| *GPT-4 + GPT4* | 80.86 | 83.44 |
| *GPT-4 + Claude 2* | **87.67** | **89.23** |
| *GPT-4 + NLI* | 79.96 | 80.16 |
| *GPT-4 + AlignScore* | 76.20 | 77.33 |
| *GPT-4 + RepC* | 79.63 | 79.23 |
| *Claude 2 + GPT4* | 79.18 | 82.89 |
| *Claude 2 + Claude 2* | 85.70 | 87.15 |
| *Claude 2 + NLI* | 76.40 | 76.29 |
| *Claude 2 + AlignScore* | 73.47 | 74.91 |
| *Claude 2 + RepC* | 76.01 | 76.48 |
| *Mistral-SFT + GPT4* | 80.98 | 83.92 |
| *Mistral-SFT + Claude 2* | 85.46 | 86.49 |
| *Mistral-SFT + NLI* | **78.54** | **79.66** |
| *Mistral-SFT + AlignScore* | 75.10 | 76.08 |
| *Mistral-SFT + RepC* | 76.59 | 76.70 |

Table 13: REFCHECKER results on the SelfCheckGPT dataset. The results of SelfCheckGPT are from their paper. We highlight the best results using proprietary LLMs with blue colors and best results results using pure open-source models with orange colors.

### D.1 Internal Knowledge Bias of Evaluated Model

In order to analyze whether the evaluated LLMs generate responses based on their own knowledge or the provided context in Noisy and Accurate Context settings, we convert each claim-triplet extracted from the response into a simple interrogative query for knowledge checking. For simplicity, we design a prompt template and ask GPT-4-Turbo to generate these queries (Figure 19). Then we feed the query into the evaluated LLM to check whether it has such knowledge. The answer from the evaluated LLMs could be one of the following:

1. `Yes`, means the evaluated LLM has this knowledge in its internal memory.
2. `No`, means the evaluated LLM contains knowledge that is contradicted with the triplet.
3. `Unsure`, means the evaluated LLM does not have this knowledge or it has confusion on the knowledge.

The label pairs (`Yes`, Contradiction) and (`Yes`, Neutral) indicate that the model is utilizing internal information to generate this claim-triplet. On the other hand, (`No`, Entailment) and (`Unsure`, Entailment) signify that the model is relying on contextual information for generation. Pairs like (`No`, Contradiction) suggest that the evaluated model may be less proficient in processing context information,